Shubham Kaushik

Data Systems Researcher | Software Engineer
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RESEARCH INTERESTS

Database Systems, Storage Engines, Data-intensive Infrastructure, and Learning-based System Tuning

EDUCATION

Doctor of Philosophy | **Brandeis University** — **Jan 2024 - Present**
Major: *Computer Science* (Current GPA: *3.98/4.0*) — Waltham, MA, USA

Master of Science | **Boston University** — **Sep 2022 - Dec 2023**
Major: *Computer Science* with specialization in *"Data-Centric Computing"* (GPA: *3.88/4.0*) — Boston, MA, USA

Bachelor of Technology | **Maharshi Dayanand University** — **Jul 2014 - Jun 2018**
Major: *Computer Science & Engineering* — Haryana, India
Thesis: "Fault Modelling of an Object-Oriented System using Colored Petri Nets"

EXPERIENCE

Ph.D. Researcher, **Smart & Scalable Data Systems Lab** @ Brandeis University, MA, USA — **Jan 2024 - Present**
- Design and evaluate LSM-based storage engine internals such as merge policy, layouts, and write-buffer design.
- Apply learning-based methods to storage-engine configuration, from offline tuning to runtime adaptation.
- Develop benchmarking suites used to evaluate key-value stores under dynamically shifting workloads.
- Built a Domain-Specific Language that restrict LLMs to generate eight edit commands to produce valid benchmarks.

Software Engineer, **Scalability & Infrastructure Team** @ Kwalee, India — **Mar 2022 - Aug 2022**

Engineer - Information Security, **Information Security Automation** @ FIS Global, India — **Jun 2021 - Mar 2022**
- Built an asynchronous IP-scanner to scan over 100K+ internet endpoints, replacing the team's manual spot-checks.
- Authored and merged integrations for DB2, GenericSQL and FireMon into Palo Alto's Cortex XSOAR [demisto/content].

Project Engineer, **Cloud Computing and Big Data Teams** @ Wipro Limited, India — **Jul 2018 - Jun 2021**
- Built a microservice that pseudonymises PII in CSV, Excel and mainframe files, replacing a manual review of 1K+ files.
- Designed a real-time agent that flags anomalous hosts across 500+ enterprise systems using k-means clustering.

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Rust
Databases: RocksDB, PostgreSQL, Redis
Systems & Performance: Linux, perf, gdb, CMake, Docker
Machine Learning: Reinforcement Learning (RL), LLMs, DSL, Retrieval-Augmented Generation (RAG)

PUBLICATIONS (**joint first author*)

7. *Mind the Buffer: Dissecting the LSM-Buffer Design Space* [PDF] [blog] [GitHub] — VLDB 2027
   Boao Chen*, **Shubham Kaushik***, and Subhadeep Sarkar
6. *TexBench: A Unified Benchmarking Suite for Shifting Workloads* (demonstration) [PDF] [blog] [GitHub] — VLDB 2026
   Abhishek Chanda*, **Shubham Kaushik***, Artem Lavrov, and Subhadeep Sarkar
   52nd International Conference on Very Large Data Bases
5. *TexBench: Harnessing LLMs for Efficient Key-Value Benchmarking* [PDF] — TPCTC 2026
   Shubham Kaushik*, Abhishek Chanda*, and Subhadeep Sarkar
   18th TPC Technology Conference on Performance Evaluation & Benchmarking
4. *RangeReduce: Query-Driven Compaction In LSM-Trees* [PDF] [talk] [GitHub] — ICDE 2026
   Shubham Kaushik, Manos Athanassoulis, and Subhadeep Sarkar
   42nd IEEE International Conference on Data Engineering
3. *Tectonic: Bridging Synthetic and Real-World Workloads for Key-Value Benchmarking* [PDF] [GitHub] — TPCTC 2025
   Alexander H. Ott*, **Shubham Kaushik***, James Chen, and Subhadeep Sarkar
   17th TPC Technology Conference on Performance Evaluation & Benchmarking
2. *Anatomy of the LSM Memory Buffer: Insights & Implications* [PDF] — DBTest 2024
   Shubham Kaushik and Subhadeep Sarkar
   In Proceedings of the International Workshop on Testing Database Systems
1. *Fault Modelling of an Object-Oriented System using CPN* [PDF] — JCSE 2019
   Shubham Kaushik and Ratneshwer
   International Journal of Computer Sciences and Engineering

– *Tectonic^{++}: A Resource-Aware Key-Value Benchmarking Suite for Shifting Workloads* — (In Review)
Boao Chen*, **Shubham Kaushik***, and Subhadeep Sarkar

ONGOING WORK

FluiDB: self-tuning storage engine [blog] 2025 – Present
Designing an LSM engine that retunes itself as workloads shift, using RL and LLM-guided retrieval to search its configuration space.

Learned-LSM Buffer: workload-aware write buffer 2026 – Present
Developing a learned memory buffer that predicts workload shifts and switches its implementation to hold a performance target.

TALKS [SELECTED]

RangeReduce: Query-Driven LSM Compactions @ MiDAS Seminar, Boston University, USA (*invited*) 04/2026
Tectonic: Bridging Synthetic and Real-World Workloads @ 2026 New England Systems Day, Harvard, USA 01/2026
Range Queries and LSM-Trees: From Foes to Friends @ Salesforce, USA (*invited*) 10/2025
Advancements in LSM Buffering: Performance and Implementations @ Brandeis University, USA 09/2024

POSTERS

Tectonic: Bridging Synthetic and Real-World Workloads @ 2026 New England Systems Day [PDF] 2026
Alexander H. Ott, **Shubham Kaushik**, Boao Chen, and Subhadeep Sarkar

RangeReduce: A Range Query Driven Compaction for LSM-Trees @ North East Database Day [PDF] 2025, 2024
Shubham Kaushik, Manos Athanassoulis, and Subhadeep Sarkar

TEACHING [ASSISTANT/GRADER]

Fundamentals of Computer Systems (COSI 131A), Brandeis University [FA '26] Fall 2026
Database Management Systems (COSI 127B), Brandeis University [SP '26] [SP '25] [SP '24] Spring 2026, 2025, 2024
Introduction to Computer Networking (COSI 128A), Brandeis University Fall 2025, 2024
Data Mechanics (DS 310), Boston University Fall/Spring 2023
Computer Networks (CS 455), Boston University Fall 2022

OUTREACH

Volunteered at *VLDB 2026*, managing the event and assisting session chairs Sep 2026
Judged and mentored at *HackMIT*, aiding teams with technical challenges Sep 2025, 2023
Mentored 4 teams, with an average of 20 participants at *BostonHacks* Nov 2022

PROFESSIONAL SERVICE

Program Committee Member, VLDB 2027 2026
Member, VLDB 2026 Shadow PC 2026
Web Chair for North East Database Day 2025 2025
External reviewer for IEEE International Conference on Big Data 2025, 2024
Member, ACM and IEEE 2024 – Present